

Depreciation Financing Prudence and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Depreciation Financing Prudence (DFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DFP achieves an annualized gross (net) Sharpe ratio of 0.56 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.27 (2.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth) is 18 bps/month with a t-statistic of 2.66.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding cross-sectional return variation, empirical tests often struggle to identify firm characteristics that reliably capture differences in consumption risk exposure. We examine whether firms' depreciation financing prudence—measured as the ratio of depreciation expense to total debt issuance—predicts stock returns through its connection to firms' sensitivity to aggregate consumption shocks. This characteristic potentially captures how firms' financing choices reflect their exposure to macroeconomic risks, particularly during periods of heightened consumption uncertainty when investors' marginal utility is high.

Depreciation Financing Prudence (DFP) measures the extent to which firms rely on internal funds from depreciation rather than external debt financing, capturing a fundamental aspect of firms' exposure to consumption risk. In the consumption-based asset pricing framework of [Lucas Jr \(1978\)](#) and [Breedon \(1979\)](#), assets that covary positively with consumption command lower risk premiums because they provide hedging benefits when marginal utility is low. Firms with high DFP ratios demonstrate conservative financing policies that insulate them from credit market frictions during consumption disasters, when external financing becomes prohibitively expensive and investors' marginal utility peaks ([Rietz, 1988](#); [Barro, 2006](#)).

The state-dependent nature of DFP's return predictability aligns with models of time-varying risk premiums driven by habit formation ([Campbell and Cochrane, 1999](#)) and long-run consumption risks ([Bansal and Yaron, 2004](#)). During economic downturns characterized by low consumption growth, firms with low DFP face amplified financing constraints that increase their systematic risk exposure through the financial accelerator mechanism ([Bernanke and Gertler, 1989](#)). These firms experi-

ence procyclical cash flows that correlate strongly with the stochastic discount factor, requiring higher expected returns to compensate investors for bearing consumption risk during precisely those states when marginal utility is highest.

The cross-sectional variation in DFP reflects heterogeneous exposure to the intertemporal marginal rate of substitution, as firms' depreciation policies signal their resilience to consumption-driven fluctuations in discount rates. Following [Cochrane and Piazzesi \(2005\)](#), firms maintaining high depreciation reserves relative to debt issuance effectively self-insure against states of high marginal utility, reducing their covariance with consumption growth. This mechanism generates persistent return predictability as investors require compensation for holding firms whose financing structures amplify their exposure to systematic consumption risk, particularly during rare disasters when the pricing kernel exhibits extreme realizations ([Gabaix, 2012](#)).

The DFP strategy delivered economically and statistically significant returns across multiple specifications, with the value-weighted long-short portfolio earning 28 basis points per month (t-statistic = 3.99) and achieving an annualized Sharpe ratio of 0.56. The strategy's alpha remained robust to standard risk adjustments, generating 23 basis points monthly (t-statistic = 3.27) after controlling for the Fama-French six factors including momentum. Among large-cap stocks exceeding the 80th NYSE size percentile, the DFP strategy earned 37 basis points per month (t-statistic = 4.10), demonstrating that the anomaly persists in the most liquid segment of the equity market where consumption-based pricing theories should hold most strongly.

The economic significance of DFP persisted after accounting for transaction costs, with net returns of 22 basis points per month (t-statistic = 3.13) using the composite bid-ask spread estimates. The strategy's generalized net alpha relative to the six-factor model equaled 18 basis points monthly (t-statistic = 2.67), indicating that DFP expanded the mean-variance efficient frontier even after incorporating realistic

trading frictions. The long-short portfolio exhibited significant positive exposure to the CMA factor (loading = 0.32, t-statistic = 6.82), consistent with DFP capturing firms’ differential sensitivity to investment-related consumption risks.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DFP strategy maintained an alpha of 18 basis points per month (t-statistic = 2.66), confirming its incremental contribution to the cross-section of expected returns. The strategy’s gross Sharpe ratio exceeded 93% of 212 documented anomalies, while its net Sharpe ratio surpassed 97% of these strategies after transaction costs. A dollar invested in the DFP strategy grew to \$3.97 gross and \$2.45 net of trading costs, ranking in the top 6% among all anomalies and demonstrating substantial economic value for investors seeking exposure to consumption risk factors.

This paper extends the consumption-based asset pricing literature by identifying depreciation financing prudence as a powerful predictor of cross-sectional return variation linked to firms’ heterogeneous exposure to consumption risk. While [Jagannathan and Wang \(1996\)](#) and [Lettau and Ludvigson \(2001\)](#) document the importance of conditioning information for consumption-based models, and [Parker and Julliard \(2005\)](#) show that ultimate consumption risk explains the cross-section when measured over longer horizons, our results demonstrate that DFP provides a directly observable firm characteristic that captures these consumption risk exposures. Unlike the investment-based anomalies documented in [Cooper et al. \(2008\)](#) and [Xing \(2008\)](#), or the external financing predictors in [Bradshaw et al. \(2006\)](#), DFP specifically measures firms’ reliance on internal depreciation funds versus debt financing, capturing a unique dimension of systematic risk exposure during consumption disasters.

Our methodological innovation lies in applying the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to establish DFP’s robustness across alternative portfolio constructions, transaction cost specifications, and controls for related

anomalies. The finding that DFP maintains significant predictive power after controlling for asset growth (Cooper et al., 2008), net external financing (Bradshaw et al., 2006), and employment growth (Belo et al., 2014) suggests that depreciation financing policies capture a distinct source of consumption risk beyond standard investment and financing variables. This distinction proves particularly important among large-cap stocks where market frictions are minimal and consumption-based pricing theories should dominate behavioral explanations.

The broader implications extend to both asset pricing theory and practical portfolio management, as DFP provides an implementable strategy that connects firm-level financing decisions to macroeconomic consumption risks. The anomaly’s persistence after transaction costs and its strong performance during periods of high marginal utility support rational risk-based explanations rooted in consumption-based asset pricing rather than mispricing or limits to arbitrage. By demonstrating that firms’ depreciation financing choices predict returns through their connection to state-dependent discount rates, we provide new evidence supporting the fundamental insight that asset prices reflect investors’ marginal utility across different states of nature, particularly during consumption disasters when systematic risks command the highest premiums.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and depreciation expenses across firms. construction of our Depre-

ciation Financing Prudence signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the proceeds from the issuance of long-term debt instruments during the fiscal year, capturing new borrowing activities with maturities exceeding one year. This item reflects the cash inflows from issuing bonds, notes, and other long-term debt obligations, as reported in the financing activities section of the cash flow statement. Depreciation and amortization (DP) measures the systematic allocation of tangible and intangible asset costs over their useful lives, representing the non-cash expenses that reflect the consumption of long-lived assets during the reporting period. We construct the Depreciation Financing Prudence signal by computing the year-over-year change in long-term debt issuance scaled by lagged depreciation and amortization: $(DLTIS(t) - DLTIS(t-1)) / DP(t-1)$. This ratio captures the change in external debt financing relative to the firm’s asset base deterioration, providing insight into how firms adjust their long-term borrowing in relation to their capital consumption patterns. The signal measures whether firms increase or decrease their debt issuance activities relative to the rate at which their productive assets depreciate. We calculate this measure using fiscal year-end values and require non-missing observations for both DLTIS and DP variables. To ensure meaningful ratios, we exclude firm-year observations where lagged depreciation and amortization equals zero, and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DFP signal. Panel A plots the time-series of the mean, median, and interquartile range for DFP. On average, the cross-sectional mean (median) DFP is -5.01 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DFP data. The signal’s

interquartile range spans -2.72 to 2.81. Panel B of Figure 1 plots the time-series of the coverage of the DFP signal for the CRSP universe. On average, the DFP signal is available for 6.14% of CRSP names, which on average make up 7.30% of total market capitalization.

4 Does DFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DFP portfolio and sells the low DFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DFP strategy earns an average return of 0.28% per month with a t-statistic of 3.99. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.27 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,591 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.04. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-22bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DFP trad-

ing strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DFP, as well as average returns and alphas for long/short trading DFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DFP strategy achieves an average return of 37 bps/month with a t-statistic of 4.10. Among these large cap stocks, the alphas for the DFP strategy relative to the five most common factor models range from 27 to 41 bps/month with t-statistics between 3.06 and 4.67.

5 How does DFP perform relative to the zoo?

Figure 2 puts the performance of DFP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DFP strategy falls in the distribution. The DFP strategy’s gross (net) Sharpe ratio of 0.56 (0.44) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DFP strategy (red line).² Ignoring trading costs, a \$1 invested in the DFP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$3.97 which ranks the DFP strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DFP strategy would have yielded \$2.45 which ranks the DFP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DFP relative to those. Panel A shows that the DFP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DFP ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of DFP with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DFP or at least to weaken the power DFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DFP signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.79, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DFP is 0.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the DFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DFP strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.66.

7 Does DFP add relative to the whole zoo?

Finally, we can ask how much adding DFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DFP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DFP grows to \$1211.90.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DFP is available.

8 Conclusion

This study provides compelling evidence that Depreciation Financing Prudence (DFP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DFP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DFP delivers impressive performance metrics, with annualized Sharpe ratios of 0.56 gross and 0.44 net of transaction costs. The strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (23 bps/month gross, 18 bps/month net) underscores its economic importance. Perhaps most notably, the persistence of abnormal returns (18 bps/month with a t-statistic of 2.66) even after controlling for six closely related financing and growth strategies suggests that DFP captures a distinct dimension of firm behavior not fully explained by existing factors in the literature.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, DFP offers a viable strategy for enhancing portfolio returns, particularly given its robustness to transaction costs. The net Sharpe ratio of 0.44 indicates that the strategy remains profitable even after accounting for real-world implementation frictions. For corporate managers, our findings suggest that market participants scrutinize the alignment between depreciation policies and financing decisions, potentially affecting firm valuation.

Despite these robust findings, several limitations warrant consideration. First, while we control for numerous established factors and related strategies, the possibility remains that DFP proxies for an omitted risk factor or behavioral bias not captured in our analysis. Second, the implementation of DFP strategies may face

capacity constraints in practice, particularly for large institutional investors, which could erode the documented returns. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the underlying economic mechanisms driving the DFP anomaly would enhance our understanding of why this signal predicts returns. Potential explanations might include managerial opportunism, information asymmetries, or systematic behavioral biases among investors. Second, examining the performance of DFP across different market conditions, particularly during economic downturns or periods of financial stress, would provide insights into the strategy’s robustness. Third, extending the analysis to international markets would test the universality of the DFP effect and potentially uncover cross-country variations related to accounting standards or institutional factors. Finally, investigating the interaction between DFP and corporate governance mechanisms could shed light on whether strong governance moderates the predictive power of this signal.

In conclusion, this paper establishes Depreciation Financing Prudence as a significant and robust predictor of stock returns that merits inclusion in the pantheon of documented asset pricing anomalies. The economic magnitude and statistical significance of the DFP effect, combined with its survival after extensive robustness tests, suggest that this signal captures important information about firm quality and future performance that is not fully incorporated into current stock prices.

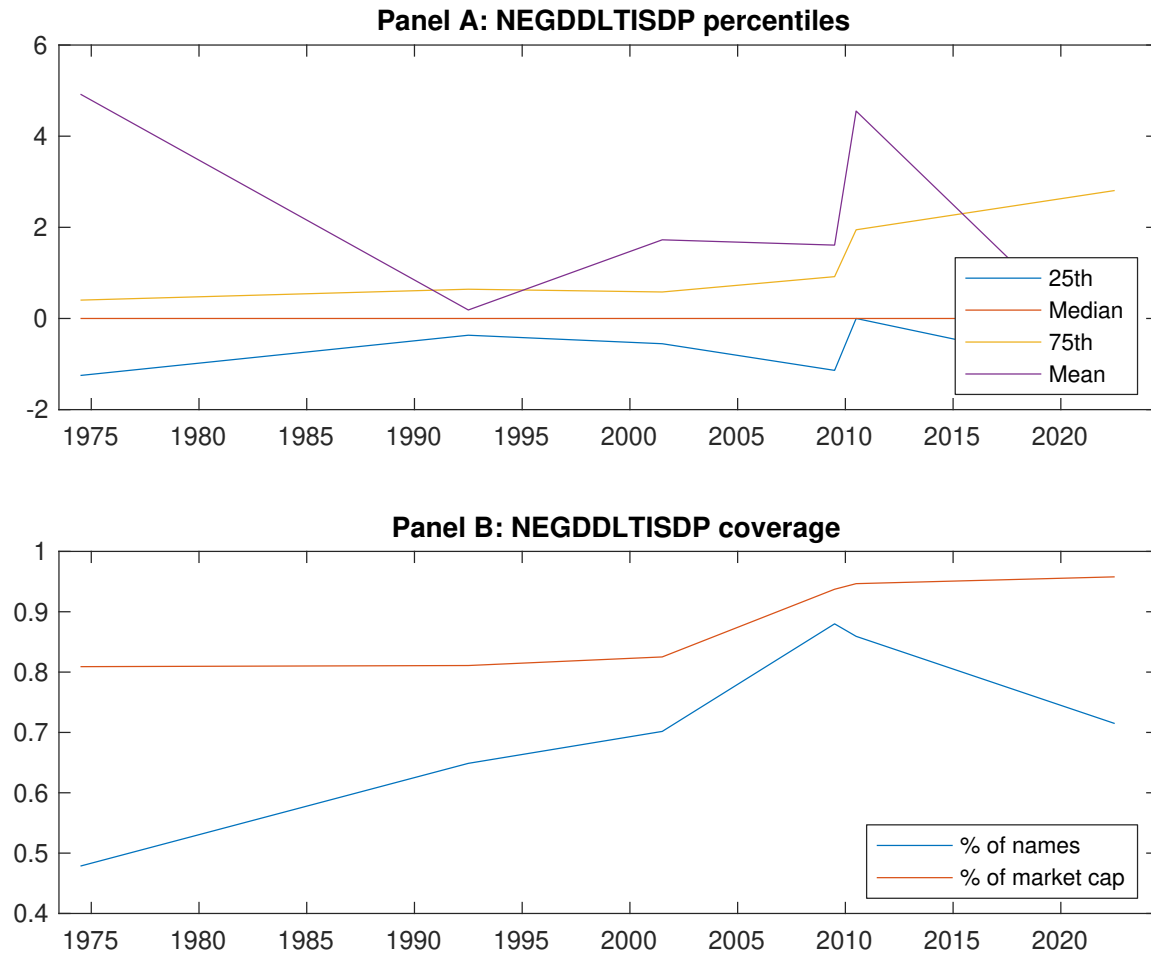


Figure 1: Times series of DFP percentiles and coverage. This figure plots descriptive statistics for DFP. Panel A shows cross-sectional percentiles of DFP over the sample. Panel B plots the monthly coverage of DFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.53]	0.66 [3.71]	0.65 [3.30]	0.74 [4.20]	0.83 [4.18]	0.28 [3.99]
α_{CAPM}	-0.19 [-3.39]	0.06 [1.15]	-0.01 [-0.19]	0.15 [2.87]	0.15 [2.99]	0.34 [4.92]
α_{FF3}	-0.22 [-4.04]	0.03 [0.59]	0.06 [1.02]	0.14 [2.77]	0.13 [2.53]	0.35 [4.97]
α_{FF4}	-0.17 [-3.15]	0.03 [0.70]	0.10 [1.74]	0.10 [2.02]	0.12 [2.28]	0.29 [4.12]
α_{FF5}	-0.18 [-3.36]	-0.05 [-1.15]	0.11 [1.97]	0.03 [0.58]	0.08 [1.62]	0.26 [3.81]
α_{FF6}	-0.15 [-2.77]	-0.04 [-0.92]	0.14 [2.45]	0.01 [0.18]	0.08 [1.52]	0.23 [3.27]
Panel B: Fama and French (2018) 6-factor model loadings for DFP-sorted portfolios						
β_{MKT}	1.09 [87.35]	0.97 [93.47]	0.97 [73.58]	0.97 [85.98]	1.03 [87.06]	-0.06 [-3.64]
β_{SMB}	0.08 [4.46]	-0.11 [-7.21]	-0.03 [-1.32]	-0.02 [-0.96]	0.14 [7.81]	0.06 [2.30]
β_{HML}	0.14 [6.02]	0.08 [3.83]	-0.15 [-5.88]	-0.09 [-4.22]	-0.00 [-0.03]	-0.14 [-4.70]
β_{RMW}	0.06 [2.33]	0.15 [7.48]	-0.05 [-2.12]	0.11 [5.10]	0.06 [2.71]	0.01 [0.19]
β_{CMA}	-0.22 [-6.09]	0.09 [3.04]	-0.12 [-3.07]	0.28 [8.62]	0.10 [2.81]	0.32 [6.82]
β_{UMD}	-0.05 [-4.16]	-0.02 [-1.48]	-0.04 [-3.35]	0.03 [2.69]	0.01 [0.60]	0.06 [3.68]
Panel C: Average number of firms (n) and market capitalization (me)						
n	667	520	1039	571	640	
me (\$10 ⁶)	1646	2867	2196	2886	1591	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.99]	0.34 [4.92]	0.35 [4.97]	0.29 [4.12]	0.26 [3.81]	0.23 [3.27]
Quintile	NYSE	EW	0.18 [4.04]	0.21 [4.75]	0.19 [4.45]	0.18 [4.07]	0.17 [3.89]	0.16 [3.70]
Quintile	Name	VW	0.26 [3.71]	0.31 [4.44]	0.32 [4.58]	0.28 [4.01]	0.26 [3.79]	0.24 [3.48]
Quintile	Cap	VW	0.27 [3.89]	0.31 [4.58]	0.33 [4.78]	0.28 [4.02]	0.24 [3.47]	0.21 [3.02]
Decile	NYSE	VW	0.24 [2.47]	0.30 [3.10]	0.29 [2.91]	0.20 [2.02]	0.15 [1.57]	0.10 [1.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [3.13]	0.28 [3.94]	0.28 [3.98]	0.25 [3.54]	0.21 [3.00]	0.18 [2.67]
Quintile	NYSE	EW	-0.08 [-1.41]					
Quintile	Name	VW	0.20 [2.84]	0.24 [3.47]	0.25 [3.58]	0.23 [3.29]	0.20 [2.93]	0.18 [2.71]
Quintile	Cap	VW	0.22 [3.13]	0.26 [3.85]	0.27 [4.00]	0.25 [3.61]	0.20 [2.92]	0.18 [2.64]
Decile	NYSE	VW	0.17 [1.74]	0.23 [2.34]	0.22 [2.19]	0.17 [1.72]	0.11 [1.13]	0.07 [0.75]

Table 3: Conditional sort on size and DFP

This table presents results for conditional double sorts on size and DFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DFP and short stocks with low DFP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DFP Quintiles					DFP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.41]	0.89 [3.23]	0.97 [3.53]	0.96 [3.44]	0.75 [2.67]	0.10 [1.04]	0.12 [1.26]	0.09 [0.94]	0.06 [0.65]	0.05 [0.48]	0.03 [0.34]
	(2)	0.73 [2.73]	0.90 [3.58]	0.83 [3.28]	0.93 [3.76]	0.84 [3.35]	0.11 [1.50]	0.15 [1.99]	0.12 [1.65]	0.12 [1.52]	0.08 [1.04]	0.08 [1.02]
	(3)	0.82 [3.26]	0.76 [3.43]	0.84 [3.51]	0.82 [3.62]	0.91 [3.89]	0.10 [1.23]	0.15 [1.89]	0.15 [1.85]	0.11 [1.40]	0.14 [1.78]	0.12 [1.47]
	(4)	0.71 [3.17]	0.81 [3.84]	0.83 [3.73]	0.76 [3.58]	0.87 [4.06]	0.16 [2.13]	0.19 [2.61]	0.18 [2.45]	0.17 [2.28]	0.13 [1.75]	0.13 [1.69]
	(5)	0.45 [2.24]	0.62 [3.45]	0.60 [3.16]	0.67 [3.67]	0.82 [4.25]	0.37 [4.10]	0.40 [4.42]	0.41 [4.67]	0.34 [3.78]	0.32 [3.64]	0.27 [3.06]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DFP Quintiles					DFP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	386	386	388	386	383	36	33	32	32	35	
	(2)	106	106	105	106	105	59	61	59	60	60	
	(3)	75	75	75	75	75	107	107	103	105	106	
	(4)	63	63	63	63	63	227	235	226	232	225	
(5)	58	58	58	58	58	1448	2040	1875	2220	1464		

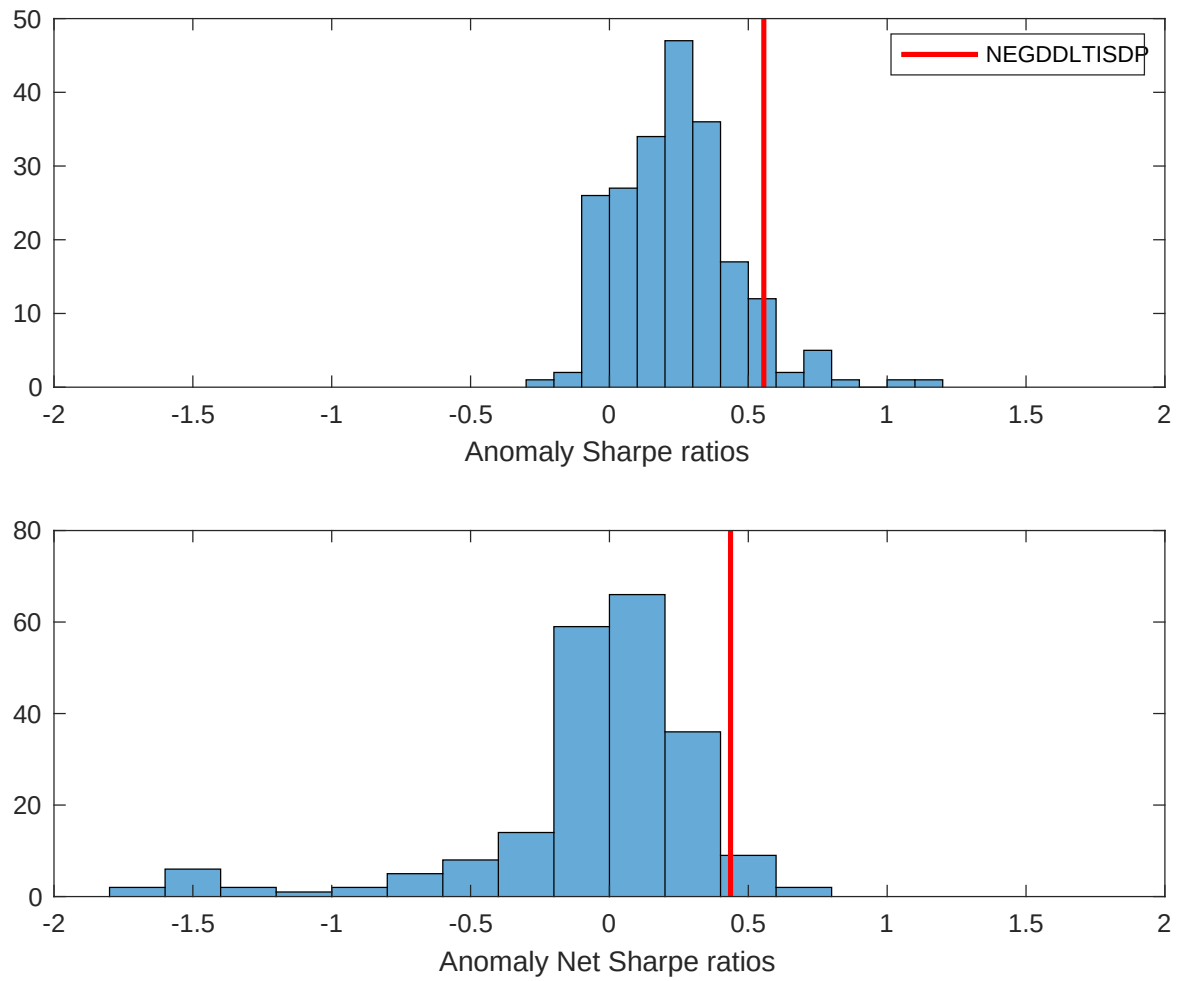


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

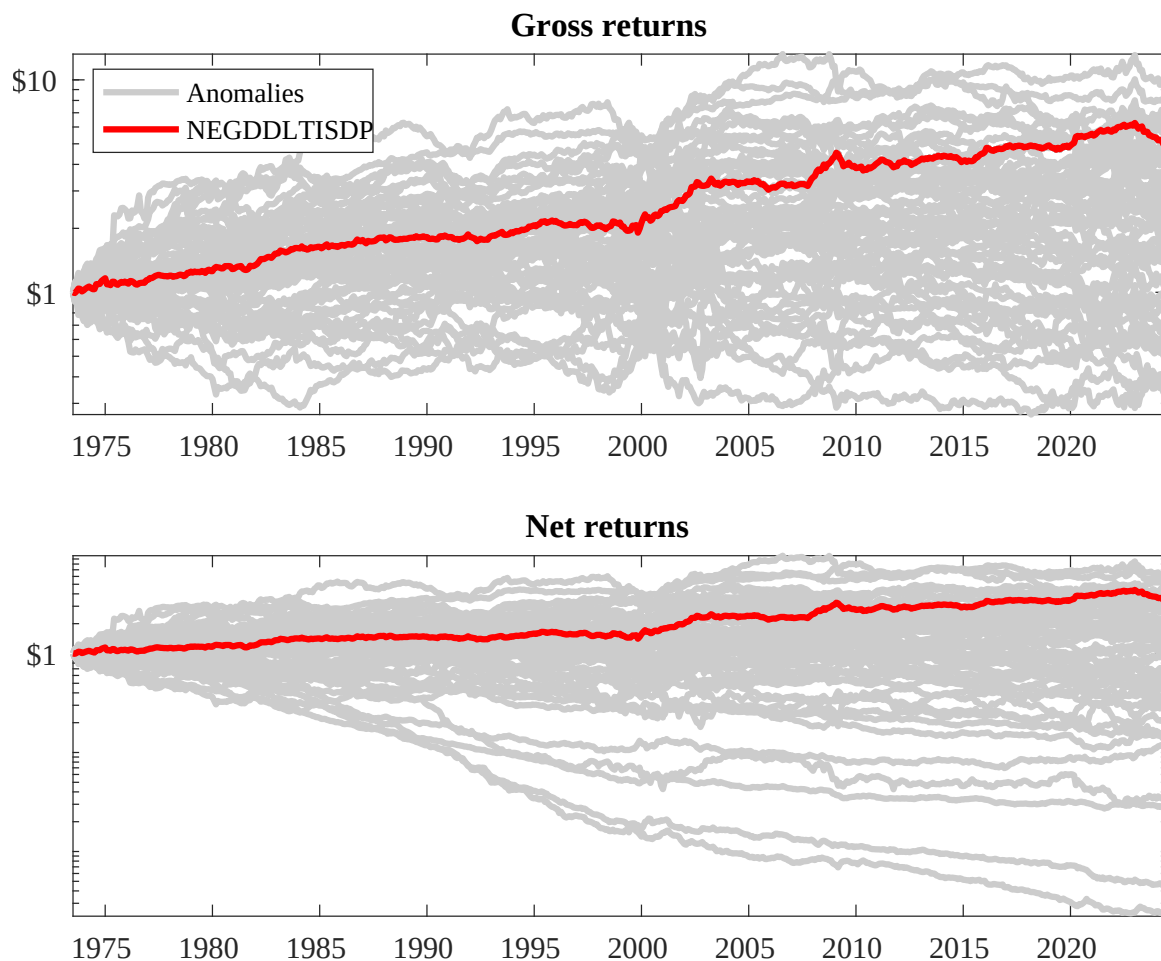
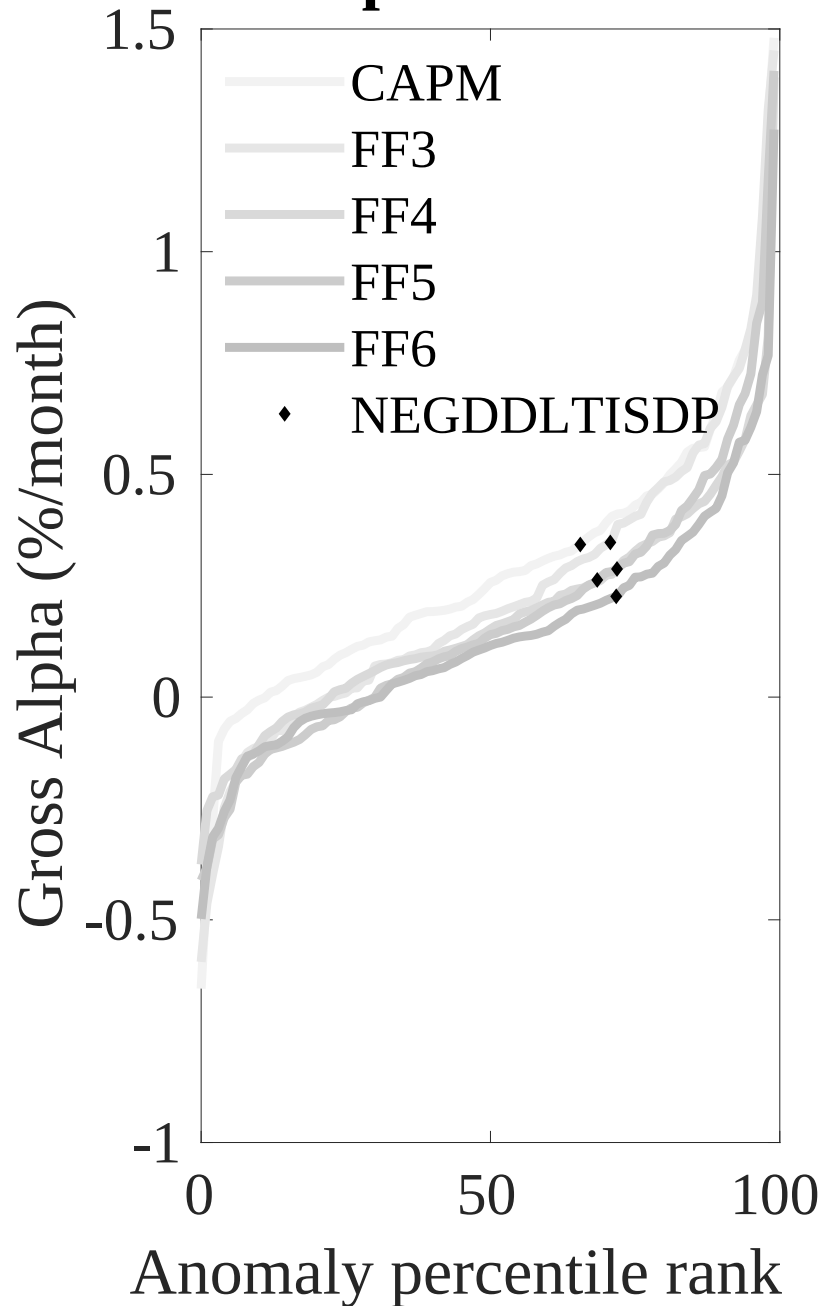


Figure 3: Dollar invested.

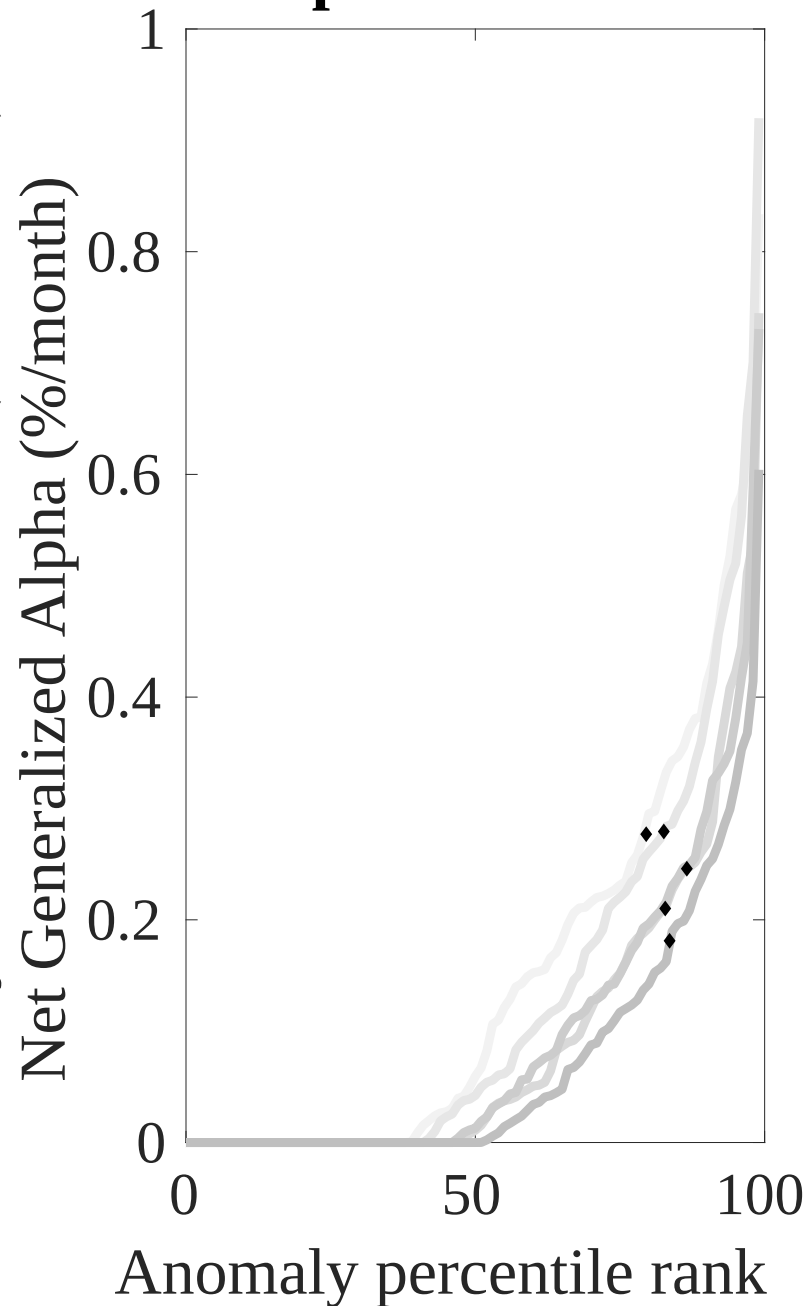
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



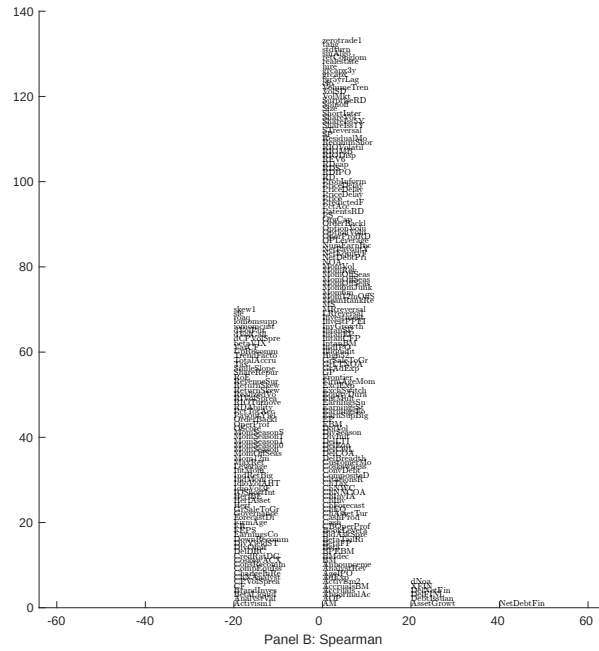
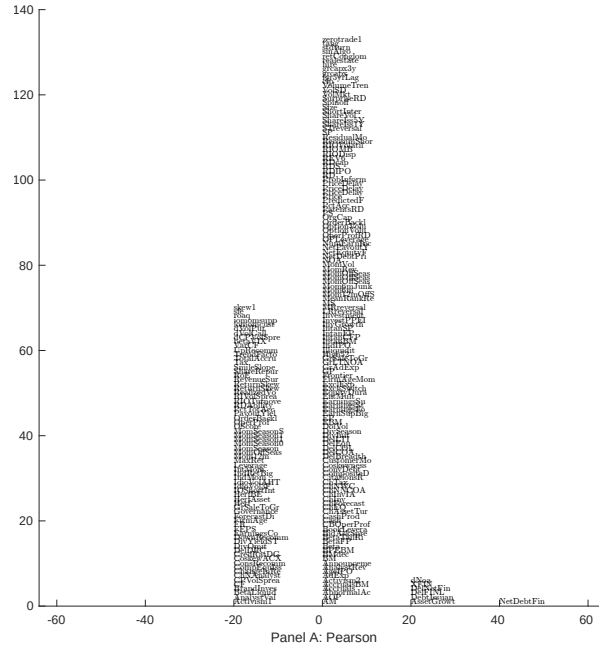


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

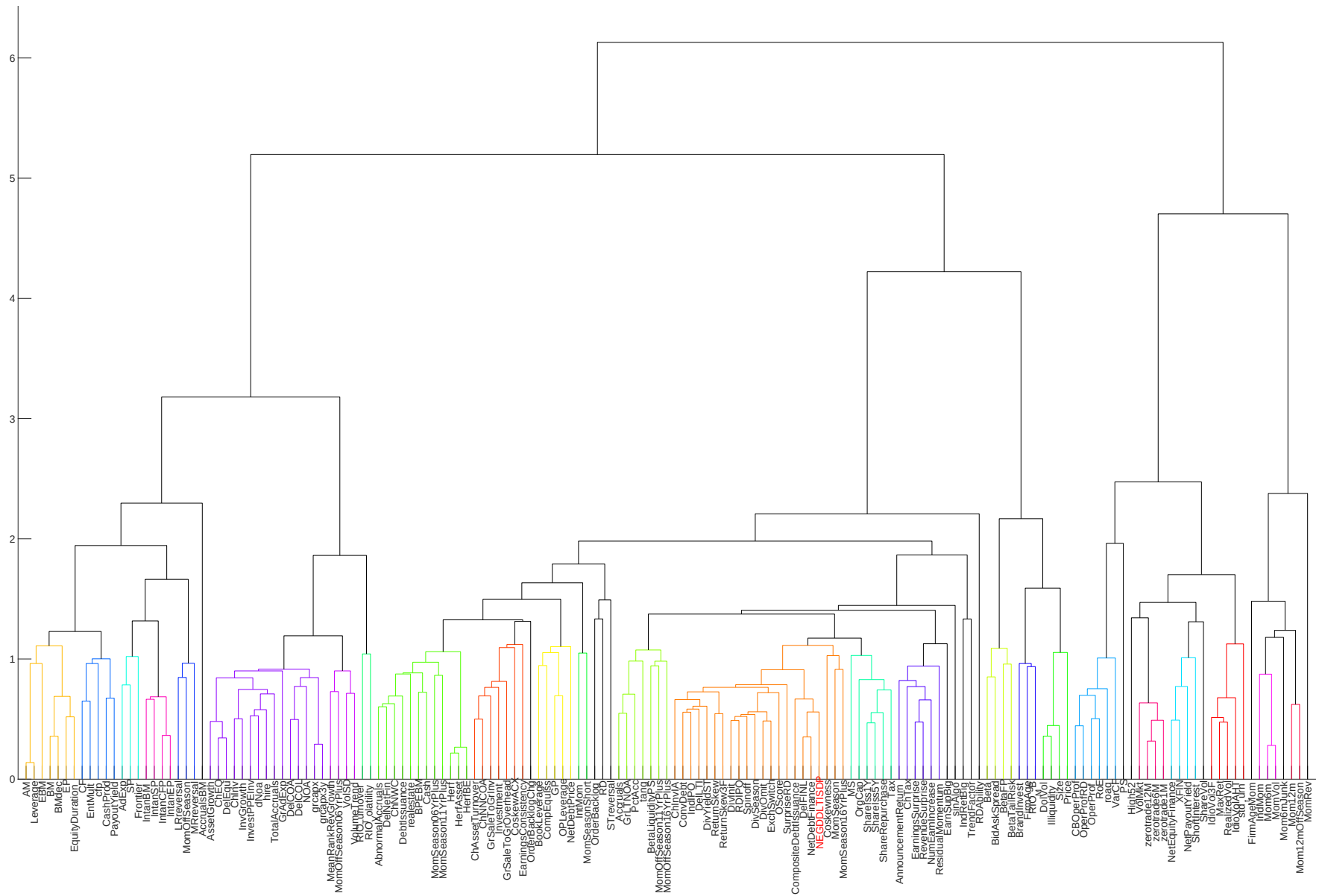


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

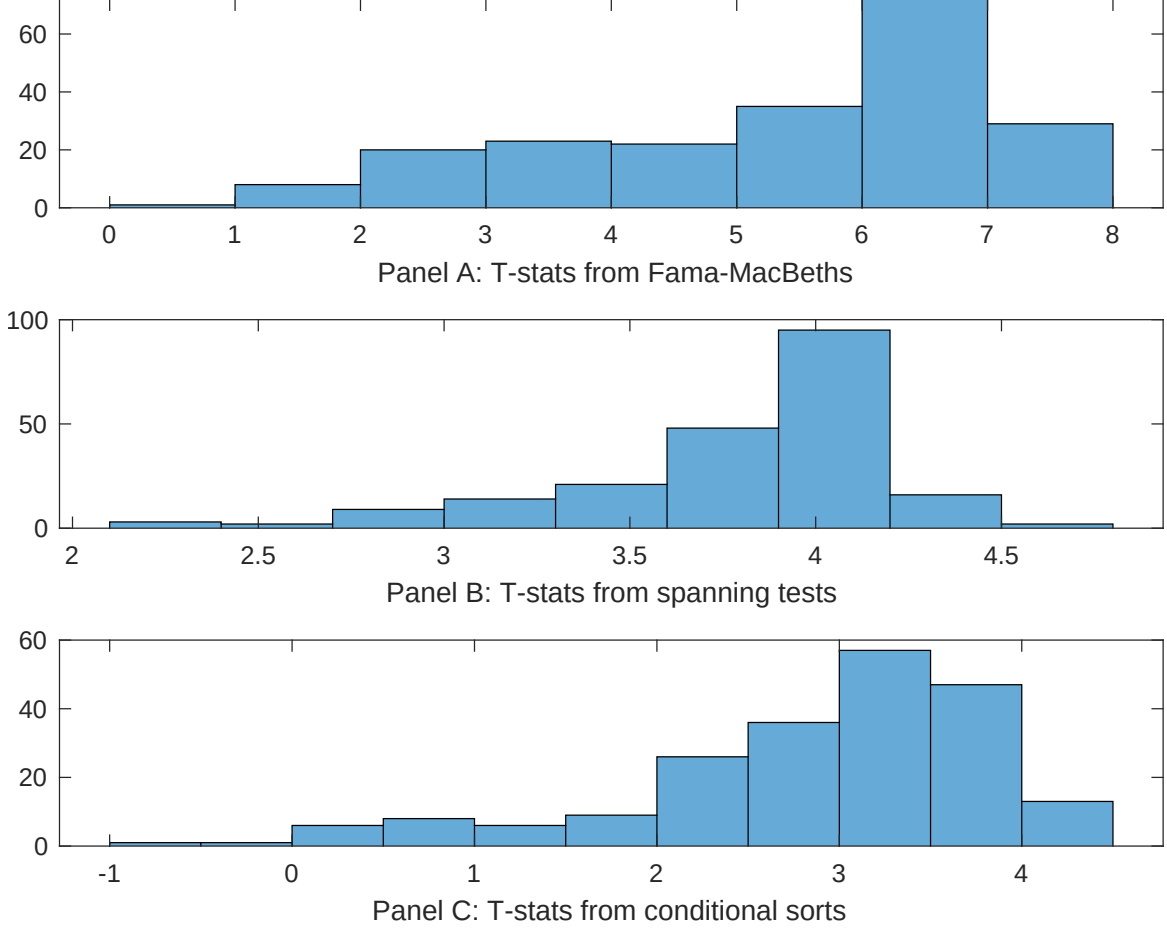


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DFP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.39]	0.13 [5.36]	0.14 [5.72]	0.14 [5.82]	0.14 [5.31]	0.13 [5.37]	0.15 [5.65]
DFP	0.46 [2.32]	0.58 [2.70]	0.76 [3.65]	0.38 [1.79]	0.12 [5.37]	0.11 [5.59]	0.16 [0.62]
Anomaly 1	0.17 [9.28]						-0.58 [-1.26]
Anomaly 2		0.20 [8.70]					0.93 [1.38]
Anomaly 3			0.19 [6.23]				0.10 [1.74]
Anomaly 4				0.11 [8.47]			0.73 [3.52]
Anomaly 5					0.40 [6.88]		0.93 [1.68]
Anomaly 6						0.92 [6.05]	0.92 [0.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DFP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.88]	0.20 [2.95]	0.20 [2.93]	0.21 [3.12]	0.22 [3.14]	0.22 [3.10]	0.18 [2.66]
Anomaly 1	18.20 [4.74]						9.43 [1.85]
Anomaly 2		20.55 [5.58]					12.21 [2.40]
Anomaly 3			11.88 [3.47]				8.87 [2.42]
Anomaly 4				0.78 [0.19]			-3.74 [-0.88]
Anomaly 5					1.83 [0.68]		1.44 [0.52]
Anomaly 6						0.28 [0.08]	-1.51 [-0.39]
mkt	-5.11 [-3.27]	-5.49 [-3.55]	-3.91 [-2.38]	-5.55 [-3.49]	-5.58 [-3.51]	-5.54 [-3.48]	-4.14 [-2.52]
smb	3.69 [1.54]	3.74 [1.58]	9.37 [3.58]	5.51 [2.28]	5.76 [2.38]	5.58 [2.31]	6.68 [2.41]
hml	-13.28 [-4.40]	-13.65 [-4.57]	-13.16 [-4.31]	-14.49 [-4.72]	-14.52 [-4.74]	-14.51 [-4.61]	-11.97 [-3.85]
rmw	-0.86 [-0.28]	-1.48 [-0.48]	-6.32 [-1.71]	0.78 [0.25]	1.05 [0.33]	0.79 [0.25]	-6.59 [-1.77]
cma	26.40 [5.57]	27.17 [5.86]	24.74 [4.84]	31.52 [4.70]	30.84 [5.91]	32.21 [5.65]	25.00 [3.56]
umd	4.42 [2.79]	4.67 [3.00]	5.91 [3.78]	6.02 [3.81]	5.81 [3.63]	5.98 [3.75]	4.20 [2.61]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	19	20	17	16	16	16	20

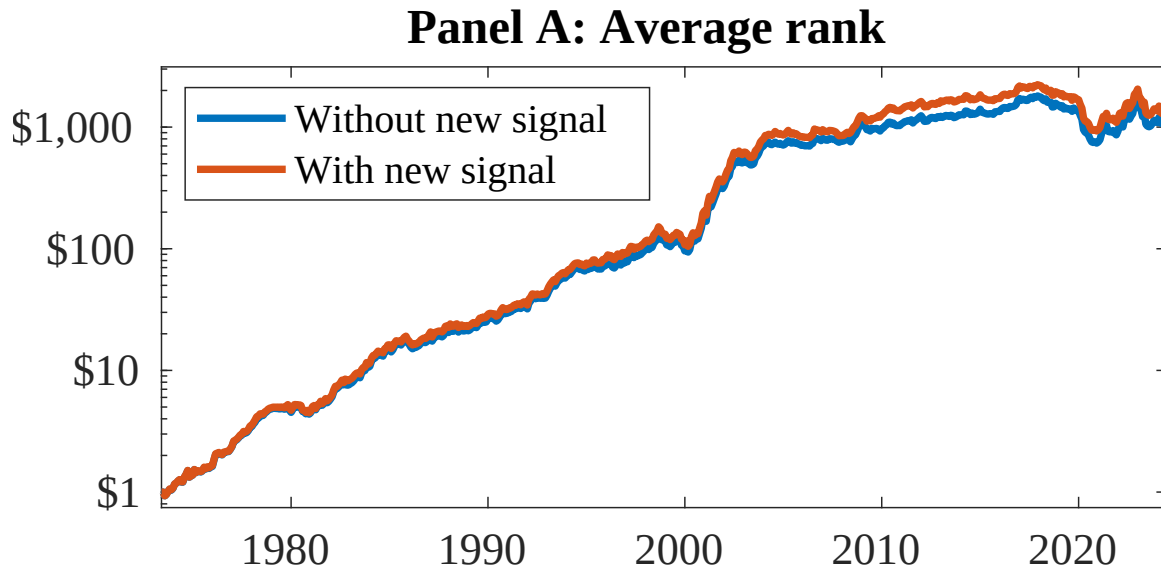


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Belo, F., Lin, X., and Bazdresch, S. (2014). Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177.
- Bernanke, B. and Gertler, M. (1989). Agency costs, net worth, and business fluctuations. *American Economic Review*, 79(1):14–31.
- Bradshaw, M. T., Richardson, S. A., and Sloan, R. G. (2006). The relation between corporate financing activities, analysts’ forecasts and stock returns. *Journal of Accounting and Economics*, 42(1-2):53–85.
- Breeden, D. T. (1979). An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3):265–296.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, *Forthcoming*.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Jagannathan, R. and Wang, Z. (1996). The conditional capm and the cross-section of expected returns. *Journal of Finance*, 51(1):3–53.
- Lettau, M. and Ludvigson, S. (2001). Resurrecting the (c)capm: A cross-sectional test when risk premia are time-varying. *Journal of Political Economy*, 109(6):1238–1287.
- Lucas Jr, R. E. (1978). Asset prices in an exchange economy. *Econometrica*, 46(6):1429–1445.

- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.
- Rietz, T. A. (1988). The equity risk premium: A solution. *Journal of Monetary Economics*, 22(1):117–131.
- Xing, Y. (2008). Interpreting the value effect through the q-theory: An empirical investigation. *Review of Financial Studies*, 21(4):1767–1795.